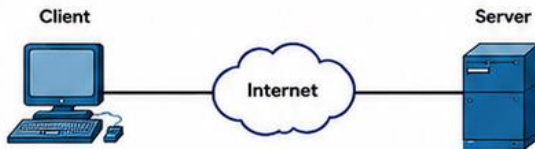


NETWORK COMPONENTS

Building Blocks of Computer Networks

1.2.1 HOST ROLES

All devices that are connected to a network and participate directly in network communication are called hosts. Hosts can be called end devices.



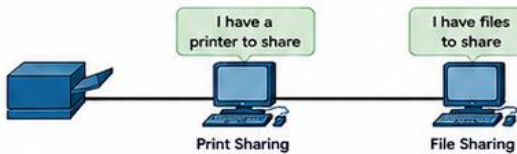
Servers are computers with software that allow them to provide information, like email or web pages, to other end devices on the network. Each service requires separate server software.

Clients are a type of host that have software for requesting and displaying the information obtained from the server.

Type	Description
Email	The email server runs email server software. Clients use mail client software, such as Microsoft Outlook, to access email on the server.
Web	The web server runs web server software. Clients use browser software, such as Windows Internet Explorer or Chrome, to access web pages on the server.
File	The file server stores corporate and user files in a central location. The client devices access these files with client software such as the Windows File Explorer.

1.2.2 PEER-TO-PEER

Client and server software usually run on separate computers, but it is also possible for one computer to be used for both roles at the same time. In small businesses and homes, many computers function as both the servers and clients on the network. This type of network is called a peer-to-peer network.



Advantages

- Easy to set up
- Less complex
- Lower cost (no dedicated servers needed)
- Can be used for simple tasks such as transferring files and sharing printers

Disadvantages

- No centralized administration
- Not as secure
- Not scalable
- All devices may act as both clients and servers which can slow performance

1.2.3 END DEVICES

The network devices that people are most familiar with are end devices. To distinguish one end device from another, each end device on a network has an address. When an end device initiates communication, it uses the address of the destination end device to specify where to deliver the message.



An end device is either the source or destination of a message transmitted over the network.

KEY TAKEAWAY

Networks are built from components that work together:

End Devices (hosts) communicate, **Intermediary Devices** connect and direct traffic, and **Network Media** carries the signals.

Understanding each component helps in designing, implementing and troubleshooting efficient networks.

1.2.4 INTERMEDIARY DEVICES

Intermediary devices connect the individual end devices to the network. They can connect multiple individual networks to form an internetwork. These devices provide connectivity and ensure that data flows across the network.

Intermediary devices use the destination end device address, in conjunction with information about the network interconnections, to determine the path that messages should take through the network.

COMMON INTERMEDIARY DEVICES

Wireless Router	LAN Switch	Router	Multilayer Switch (Layer 3 Switch)	Firewall Appliance
• Connects wireless devices to the network • Provides Internet access • Often includes firewall & DHCP	• Connects devices in a LAN • Forwards data based on MAC addresses • Each port is its own collision domain	• Connects different networks • Forwards data based on IP addresses • Chooses best path to destination	• Combines switching and routing • Works at Layer 2 and Layer 3 • High-speed routing between VLANs	• Monitors and controls incoming and outgoing network traffic • Protects network from unauthorized access • Enforces security policies

INTERMEDIARY DEVICES PERFORM THESE FUNCTIONS:

Regenerate and retransmit signals	Maintain information about network paths	Notify other devices of errors & failures	Direct data along alternate paths	Classify and prioritize messages	Permit or deny flow of data
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Note: Not shown is a legacy Ethernet hub. An Ethernet hub is also known as a multiport repeater. Repeater regenerate and retransmit communication signals. Notice that all intermediary devices perform the function of a repeater.

1.2.5 NETWORK MEDIA

Communication transmits across a network on media. The media provides the channel over which the message travels from source to destination. Modern networks primarily use three types of media to interconnect devices.

TYPE OF MEDIA	DESCRIPTION	EXAMPLES / CONNECTORS
Metal Wires within Cables	Data is encoded into electrical impulses.	
Glass or Plastic Fibers within Cables (Fiber-optic Cable)	Data is encoded into pulses of light.	
Wireless Transmission	Data is encoded via modulation of specific frequencies of electromagnetic waves.	

FOUR MAIN CRITERIA FOR CHOOSING NETWORK MEDIA

What is the maximum distance that the media can successfully carry a signal?	What is the environment in which the media will be installed?	What is the amount of data and at what speed must it be transmitted?	What is the cost of the media and installation?
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LEGEND



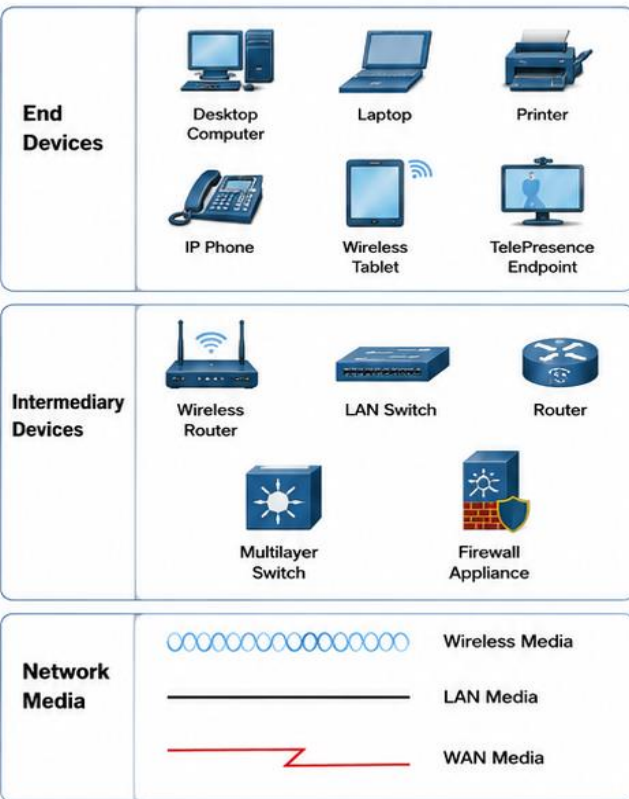


NETWORK REPRESENTATIONS AND TOPOLOGIES



1.3.1 NETWORK REPRESENTATIONS

Network architects and administrators must be able to show what their networks will look like. They need to be able to easily see which components connect to other components, where they will be located, and how they will be connected. Diagrams of networks often use symbols, like those shown in the figure, to represent the different devices and connectors that make up a network.



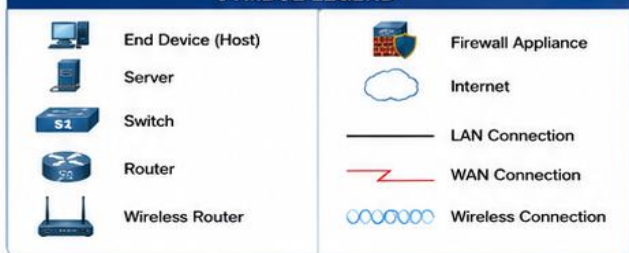
A diagram provides an easy way to understand how devices connect in a large network. This type of "picture" of a network is known as a topology diagram. The ability to recognize the logical representations of the physical networking components is critical to being able to visualize the organization and operation of a network.

In addition to these representations, specialized terminology is used to describe how each of these devices and media connect to each other:

- **Network Interface Card (NIC)** – A NIC physically connects the end device to the network.
- **Physical Port** – A connector or outlet on a networking device where the media connects to an end device or another networking device.
- **Interface** – Specialized ports on a networking device that connect to individual networks. Because routers connect networks, the ports on a router are referred to as network interfaces.

Note: The terms port and interface are often used interchangeably.

SYMBOL LEGEND

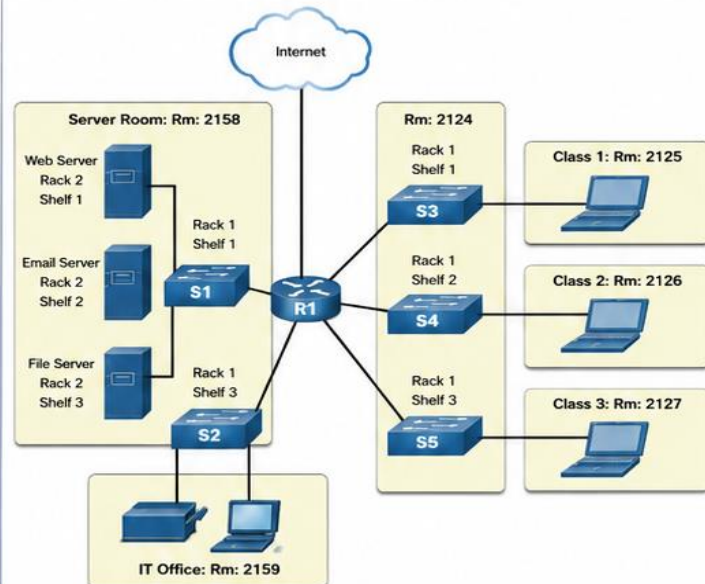


1.3.2 TOPOLOGY DIAGRAMS

Topology diagrams are mandatory documentation for anyone working with a network. They provide a visual map of how the network is connected. There are two types of topology diagrams: physical and logical.

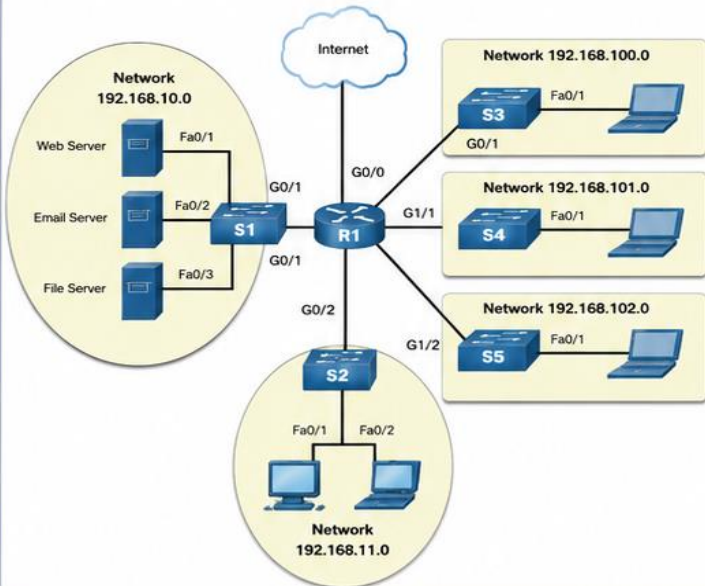
PHYSICAL TOPOLOGY DIAGRAMS

Physical topology diagrams illustrate the physical location of intermediary devices and cable installation, as shown in the figure. You can see that the rooms in which these devices are located are labeled in this physical topology.



LOGICAL TOPOLOGY DIAGRAMS

Logical topology diagrams illustrate devices, ports, and the addressing scheme of the network, as shown in the figure. You can see which end devices are connected to which intermediary devices and what media is being used.



The topologies shown in the physical and logical diagrams are appropriate for your level of understanding at this point in the course. Search the Internet for "network topology diagrams" to see some more complex examples. If you add the word "Cisco" to your search phrase, you will find many topologies using icons that are similar to what you have seen in these figures.



COMMON TYPES OF NETWORK (PART 1)



1.4.1 Networks of Many Sizes



Now that you are familiar with the components that make up networks and their representations in physical and logical topologies, you are ready to learn about the many different types of networks.



Networks come in all sizes. They range from simple networks consisting of two computers, to networks connecting millions of devices.



Simple home networks let you share resources, such as printers, documents, pictures, and music, among a few local end devices.



Small office and home office (SOHO) networks allow people to work from home, or a remote office. Many self-employed workers use these types of networks to advertise and sell products, order supplies, and communicate with customers.



Businesses and large organizations use networks to provide consolidation, storage, and access to information on network servers. Networks provide email, instant messaging, and collaboration among employees. Many organizations use their network's connection to the Internet to provide products and services to customers.



The Internet is the largest network in existence. In fact, the term Internet means a "network of networks". It is a collection of interconnected private and public networks.



In small businesses and homes, many computers function as both the servers and clients on the network. This type of network is called a peer-to-peer network.

Small Home Networks

Small Office and Home Office Networks

Medium to Large Networks

World Wide Networks



World Wide Networks

The internet is a network of networks that connects hundreds of millions of computers world-wide.

Small Home Networks

Small Office and Home Office Networks

Medium to Large Networks

World Wide Networks



Small Office and Home Office Networks

The SOHO network allows computers in a home office or a remote office to connect to a corporate network, or access centralized, shared resources.

Small Home Networks

Small Office and Home Office Networks

Medium to Large Networks

World Wide Networks



Medium to Large Networks

Medium to large networks, such as those used by corporations and schools, can have many locations with hundreds or thousands of interconnected hosts.

Small Home Networks

Small Office and Home Office Networks

Medium to Large Networks

World Wide Networks



Small Home Networks

Simple home networks let you share resources, such as printers, documents, pictures, and music, among a few local end devices.



COMMON TYPES OF NETWORK (PART 2)



1.4.2 LANs and WANs

Network infrastructures vary greatly in terms of:

- Size of the area covered
- Number of users connected
- Number and types of services available
- Area of responsibility

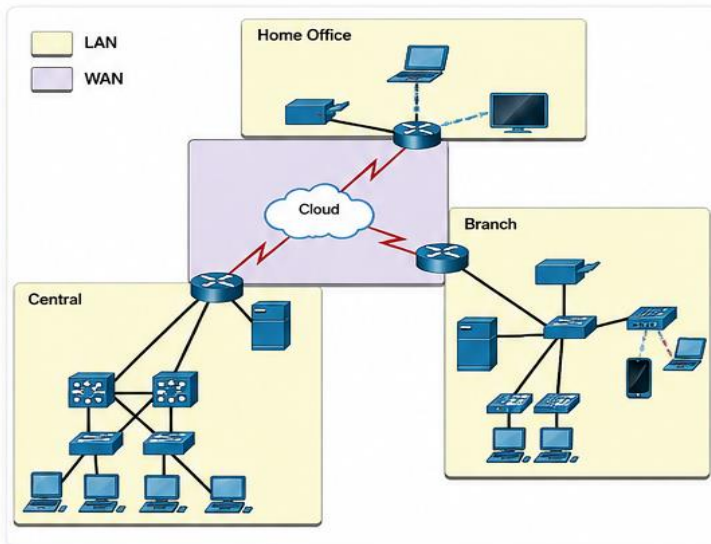
The two most common types of network infrastructures are Local Area Networks (LANs), and Wide Area Networks (WANs).

A LAN is a network infrastructure that provides access to users and end devices in a small geographical area.

A LAN is typically used in a department within an enterprise, a home, or a small business network.

A WAN is a network infrastructure that provides access to other networks over a wide geographical area, which is typically owned and managed by a larger corporation or a telecommunications service provider.

The figure shows LANs connected to a WAN.

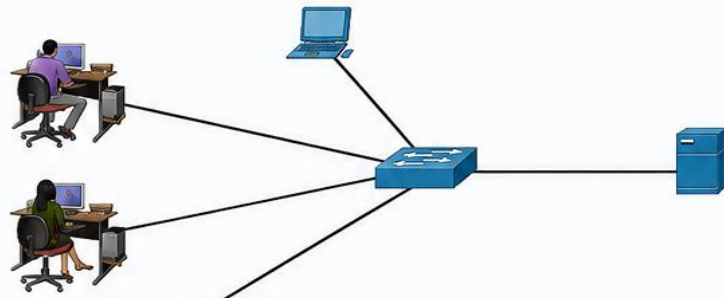


LANs

A LAN is a network infrastructure that spans a small geographical area.

LANs have specific characteristics:

- LANs interconnect end devices in a limited area such as a home, school, office building, or campus.
- A LAN is usually administered by a single organization or individual. Administrative control is enforced at the network level and governs the security and access control policies.
- LANs provide high-speed bandwidth to internal end devices and intermediary devices, as shown in the figure.



A network serving a home, small building, or a small campus is considered a LAN.

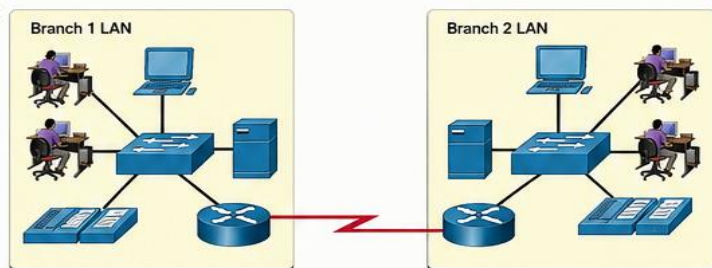
WANs

The figure shows a WAN which interconnects two LANs.

A WAN is a network infrastructure that spans a wide geographical area. WANs are typically managed by service providers (SP) or Internet Service Providers (ISP).

WANs have specific characteristics:

- WANs interconnect LANs over wide geographical areas such as between cities, states, provinces, countries, or continents.
- WANs are usually administered by multiple service providers.
- WANs typically provide lower speed links between LANs.



LAN



- Covers a small geographical area (home, school, office, campus, department)
- Managed by a single organization/individual
- High-speed bandwidth
- Provides access to local resources and end devices
- Examples: Home LAN, Office LAN, Campus LAN

Examples of LAN Devices



WAN



- Covers a wide geographical area (cities, countries, continents)
- Managed by multiple service providers (ISP)
- Lower speed links compared to LAN
- Connects multiple LANs over long distances
- Examples: Internet, Corporate WAN, Bank WAN

Examples of WAN Devices



LANs connect devices within a small area. WANs connect multiple LANs across large geographical areas.



COMMON TYPES OF NETWORK (PART 3)



1.4.3 The Internet

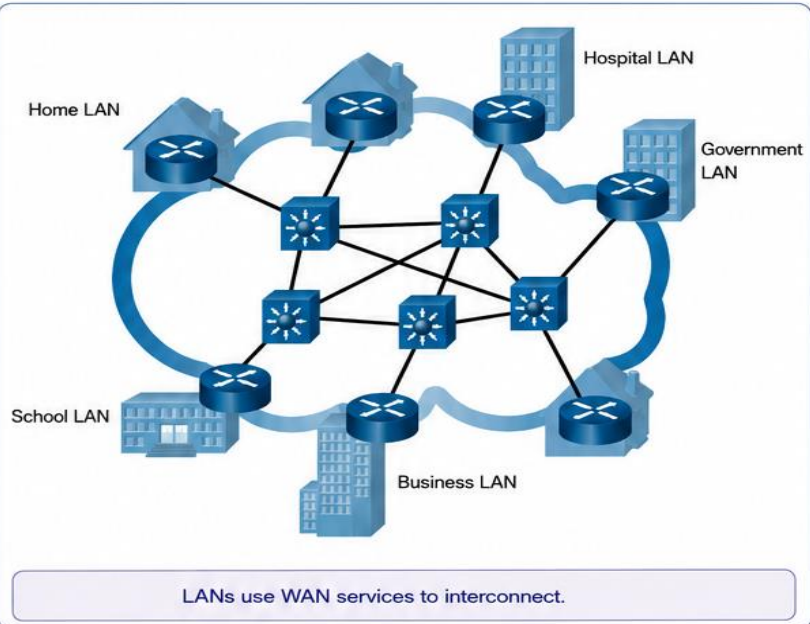
The internet is a worldwide collection of interconnected networks (internetworks, or internet for short). The figure shows one way to view the internet as a collection of interconnected LANs and WANs.

Key Points

- ✓ Some of the LAN examples are connected to each other through a WAN connection. WANs are then connected to each other.
- ✓ WANs can connect through copper wires, fiber-optic cables, and wireless transmissions (not shown).

Important Notes

- ✓ The internet is not owned by any individual or group.
- ✓ Ensuring effective communication across this diverse infrastructure requires the application of consistent and commonly recognized technologies and standards as well as the cooperation of many network administration agencies.



Organizations that help maintain the structure and standardization of internet protocols and processes:



IETF
Internet Engineering Task Force



ICANN
Internet Corporation for Assigned Names and Numbers



IAB
Internet Architecture Board



and many others.

1.4.4 Intranets and Extranets

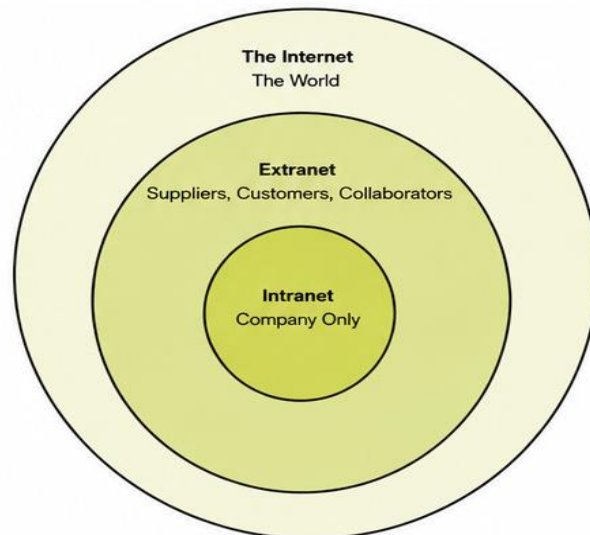
There are two other terms which are similar to the term internet: intranet and extranet.

Intranet is a term often used to refer to a private connection of LANs and WANs that belongs to an organization. An intranet is designed to be accessible only by the organization's members, employees, or others with authorization.

An organization may use an **extranet** to provide secure and safe access to individuals who work for a different organization but require access to the organization's data. Here are some examples of extranets:

- A company that is providing access to outside suppliers and contractors
- A hospital that is providing a booking system to doctors so they can make appointments for their patients
- A local office of education that is providing budget and personnel information to the schools in its district

The figure illustrates the levels of access that different groups have to a company intranet, a company extranet, and the internet.



Key Takeaway

Intranet is for internal use within an organization. Extranet extends controlled access to trusted external users.

The Internet is the global network of networks accessible to everyone.



Intranet
(Company Only)



Extranet
(Trusted External Users)



The Internet
(The World)



INTERNET CONNECTIONS (PART 1)



1.5.1 Internet Access Technologies

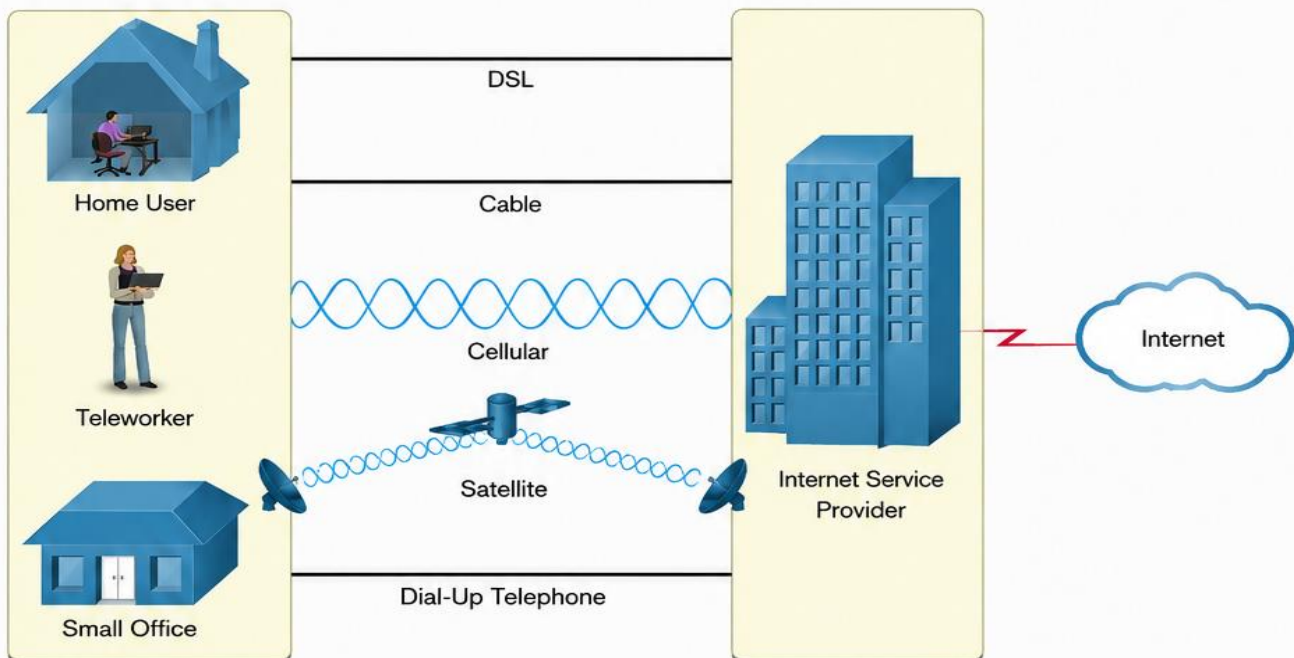
So, now you have basic understanding of what a local area network (LAN) and types of networks. But, how do you obtain or connect users and organizations to the Internet? As you may know, access comes in many different ways and cost.

Some users, remote workers, and small offices typically connect to the Internet over DSL, cable modems, or other wireless options very quickly. Modern ISPs sell gigabits of bandwidth. However, people in remote areas or broadband-deficit, rural areas may be stuck with low speed, which counts, and radio connections.

Organizations such as small schools or others connect other as well as the Internet. Such connections are regulated by a special network equipment made by numerous telecommunications, and data traffic devices. ISP's also lease or own interconnect devices. Typical business class services include leased DSL, leased lines, and Metro Ethernet.

1.5.2 Home and Small Office Internet Connections

The figure illustrates common connection options for small office and home office users.



- **Cable** – Typically offered by cable television service providers, the Internet data signal transmits on the same cable that delivers cable television. It provides a high bandwidth, high availability, and an always-on connection to the Internet.
- **DSL** – Digital Subscriber Lines also provide high bandwidth, high availability, and an always-on connection to the Internet. DSL runs over a telephone line. In general, small office and home office users connect using Asymmetrical DSL (ADSL), which means that the download speed is faster than the upload speed.
- **Cellular** – Cellular Internet access uses a cell phone network to connect. Wherever you can get a cellular signal, you can get cellular Internet access. Performance is limited by the capabilities of the phone and the cell tower to which it is connected.
- **Satellite** – The availability of satellite Internet access is a benefit in those areas that would otherwise have no Internet connectivity at all. Satellite dishes require a clear line of sight to the satellite.
- **Dial-up Telephone** – An inexpensive option that uses any phone line and a modem. The low bandwidth provided by a dial-up modem connection is not sufficient for large data transfer, although it is useful for mobile access while traveling.



The choice of connection varies depending on geographical location and service provider availability.



INTERNET CONNECTIONS (PART 1)



1.5.1 Internet Access Technologies

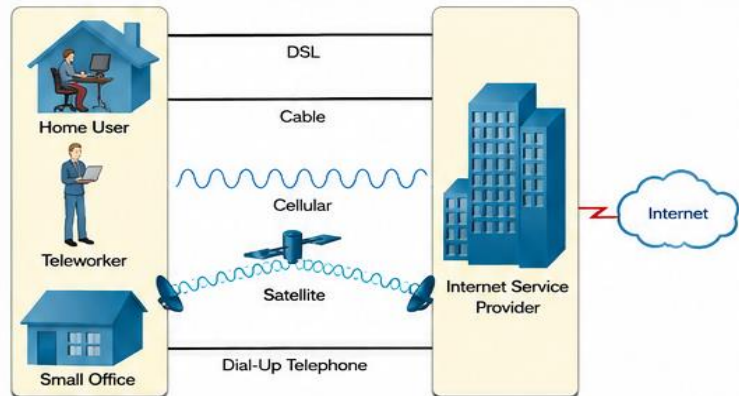
So, now you have a basic understanding of what makes up a network and the different types of networks. But, how do you actually connect users and organizations to the internet? As you may have guessed, there are many different ways to do this.

Home users, remote workers, and small offices typically require a connection to an ISP to access the internet. Connection options vary greatly between ISPs and geographical locations. However, popular choices include broadband cable, broadband digital subscriber line (DSL), wireless WANs, and mobile services.

Organizations usually need access to other corporate sites as well as the internet. Fast connections are required to support business services including IP phones, video conferencing, and data center storage. ISPs offer business-class interconnections. Popular business-class services include business DSL, leased lines, and Metro Ethernet.

1.5.2 Home and Small Office Internet Connections

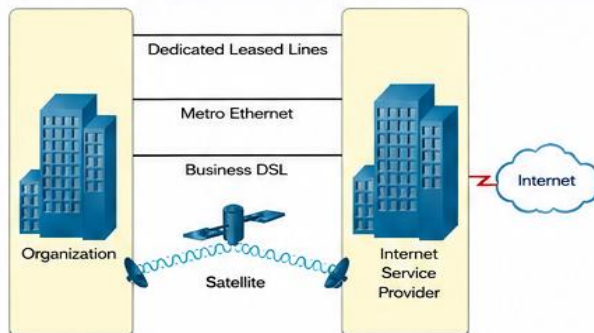
The figure illustrates common connection options for small office and home office users.



1.5.3 Business Internet Connections

Corporate connection options differ from home user options. Businesses may require higher bandwidth, dedicated bandwidth, and managed services. Connection options that are available differ depending on the type of service providers located nearby.

The figure illustrates common connection options for businesses.



- **Dedicated Leased Line** – Leased lines are reserved circuits within the service provider's network that connect geographically separated offices for private voice and/or data networking. The circuits are rented at a monthly or yearly rate.
- **Metro Ethernet** – This is sometimes known as Ethernet WAN. In this module, we will refer to it as Metro Ethernet. Metro ethernet extends LAN access technology into the WAN. Ethernet is a LAN technology you will learn about in a later module.
- **Business DSL** – Business DSL is available in various formats. A popular choice is Symmetric Digital Subscriber Line (SDSL) which is similar to the consumer version of DSL but provides uploads and downloads at the same high speeds.
- **Satellite** – Satellite service can provide a connection when a wired solution is not available.



The choice of connection varies depending on geographical location and service provider availability.

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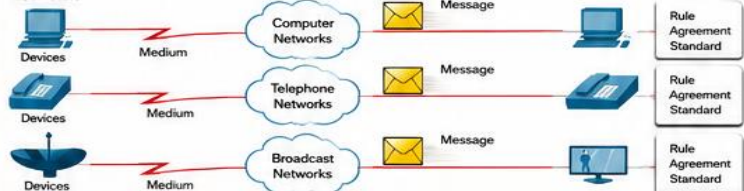


The choice of connection varies depending on geographical location and service provider availability.

1.5.4 The Converging Network

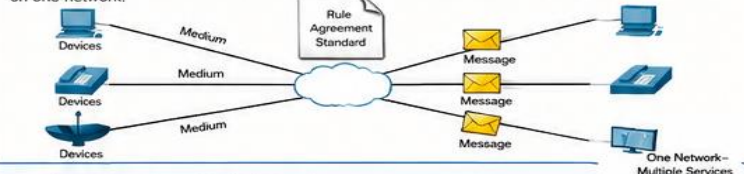
Traditional Separate Networks

Consider a school built thirty years ago. Back then, some classrooms were cabled for the data network, telephone network, and video network for televisions. These separate networks could not communicate with each other. Each network used different technologies to carry the communication signal. Each network had its own set of rules and standards to ensure successful communication. Multiple services ran on multiple networks.



Converged Networks

Today, we operate data, telephony, and video networks converge. Unlike dedicated networks, converged networks are capable of delivering data, voice, and video between many different types of devices over the same network infrastructure. This network infrastructure uses the same set of rules, agreements, and implementation standards. Converged one networks carry multiple services on one network.



Key Takeaways

- Internet access technologies provide many ways for users to connect to the Internet.
- Home and small offices commonly use DSL, cable, cellular, satellite, or dial-up connections.
- Businesses require higher performance and may use dedicated leased lines, Metro Ethernet, business DSL, or satellite.
- Networks are converging – data, voice, and video now run on a single network using the same rules and standards.





RELIABLE NETWORKS (PART 1)

Building networks that stay strong, grow easily, and serve users well!



1.6.1 Network Architecture



Have you ever been busy working online, only to have "the internet go down"? As you know by now, the internet did not go down, you just lost your connection to it. It is very frustrating.



With so many people in the world relying on network access to work and learn, it is imperative that networks are reliable. In this context, reliability means more than your connection to the internet. This topic focuses on the four aspects of network reliability.



The role of the network has changed from a data-only network to a system that enables the connections of people, devices, and information in a media-rich, converged network environment. For networks to function efficiently and grow in this type of environment, the network must be built upon a standard network architecture.



Networks also support a wide range of applications and services. They must operate over many different types of cables and devices, which make up the physical infrastructure. The term network architecture, in this context, refers to the technologies that support the infrastructure and the programmed services and rules, or protocols, that move data across the network.



As networks evolve, we have learned that there are four basic characteristics that network architects must address to meet user expectations:

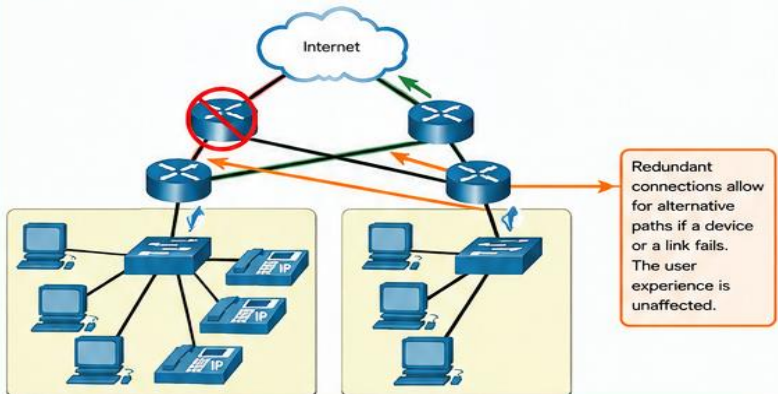
- 1 Fault Tolerance
- 2 Scalability
- 3 Quality of Service (QoS)
- 4 Security

1.6.2 Fault Tolerance

A fault tolerant network is one that limits the number of affected devices during a failure. It is built to allow quick recovery when such a failure occurs. These networks depend on multiple paths between the source and destination of a message. If one path fails, the messages are instantly sent over a different link. Having multiple paths to a destination is known as redundancy.

Implementing a packet-switched network is one way that reliable networks provide redundancy. Packet switching splits traffic into packets that are routed over a shared network. A single message, such as an email or a video stream, is broken into multiple message blocks, called packets. Each packet has the necessary addressing information of the source and destination of the message.

The routers within the network switch the packets based on the condition of the network at that moment. This means that all the packets in a single message could take very different paths to the same destination. In the figure, the user is unaware and unaffected by the router that is dynamically changing the route when a link fails.



- **Dedicated Leased Line** – Leased lines are reserved circuits within the service provider's network that connect geographically separated offices for private voice and/or data networking. The circuits are rented at a monthly or yearly rate.
- **Metro Ethernet** – This is sometimes known as Ethernet WAN. In this module, we will refer to it as Metro Ethernet. Metro ethernet extends LAN access technology into the WAN. Ethernet is a LAN technology you will learn about in a later module.
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The choice of connection varies depending on geographical location and service provider availability.

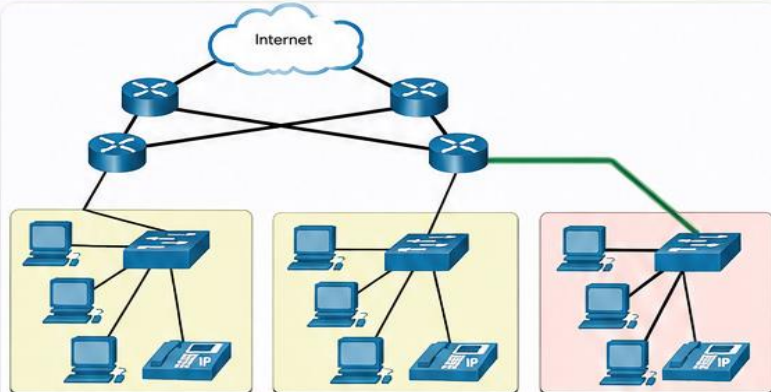
1.6.3 Scalability

A scalable network expands quickly to support new users and applications. It does this without degrading the performance of services that are being accessed by existing users.

The figure shows how a new network is easily added to an existing network. These networks are scalable because the designers follow accepted standards and protocols. This lets software and hardware vendors focus on improving products and services without having to design a new set of rules for operating within the network.



Scalability = Grow easily without slowing down!



Additional users and whole networks can be connected to the internet without degrading performance for existing users.



Key Takeaways

- ✓ Network architecture ensures networks are reliable and support people, devices, and information.
- ✓ Reliable networks must be Fault Tolerant, Scalable, provide Quality of Service (QoS), and be Secure.
- ✓ Fault tolerance uses multiple paths (redundancy) so that if one path fails, another path delivers the message.
- ✓ Scalability allows networks to grow and add new users, devices, and applications without slowing down.





RELIABLE NETWORKS (PART 2)

Four Key Characteristics of Network Reliability



1.6.4 Quality of Service (QoS)

Quality of Service (QoS) is an increasing requirement of networks today. New applications available to users over networks, such as voice and live video transmissions, create higher expectations for the quality of the delivered services.

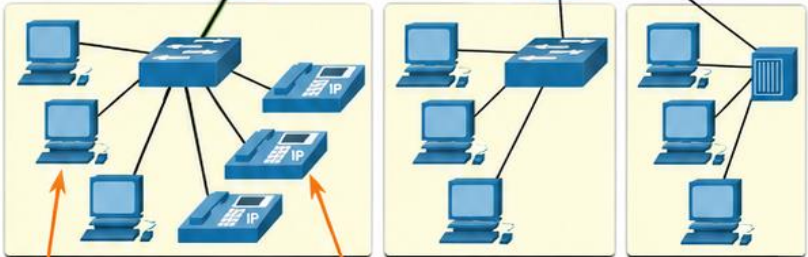
Have you ever tried to watch a video with constant breaks and pauses? As data, voice, and video content continue to converge onto the same network, QoS becomes a primary mechanism for managing congestion and ensuring reliable delivery of content to all users.

Congestion – occurs when the demand for bandwidth exceeds the amount available. Network bandwidth is measured in the number of bits that can be transmitted in a single second, or bits per second (bps). When simultaneous communications are attempted across the network, the demand for network bandwidth can exceed its availability, creating network congestion.

When the volume of traffic is greater than what can be transported across the network, devices will hold the packets in memory until resources become available to transmit them. In the figure, one user is requesting a web page, and another is on a phone call. With a QoS policy in place, the router can manage the flow of data and voice traffic, giving priority to voice communications if the network experiences congestion.

The focus of QoS is to prioritize time-sensitive traffic. The type of traffic, not the content of the traffic, is what is important.

Quality of Service, managed by the router, ensures that priorities are matched with the type of communication and its importance to the organization.



Web pages can usually receive a lower priority.



A Voice over IP (VoIP) Call will need priority to maintain a smooth, uninterrupted user experience.



KEY POINTS – QUALITY OF SERVICE

- ✓ QoS helps manage network congestion and ensures important traffic gets priority.
- ✓ It focuses on the type of traffic (time-sensitive or critical), not the content.
- ✓ Examples of time-sensitive traffic: voice calls, video conferencing, real-time gaming, live video streaming.

1.6.5 Network Security

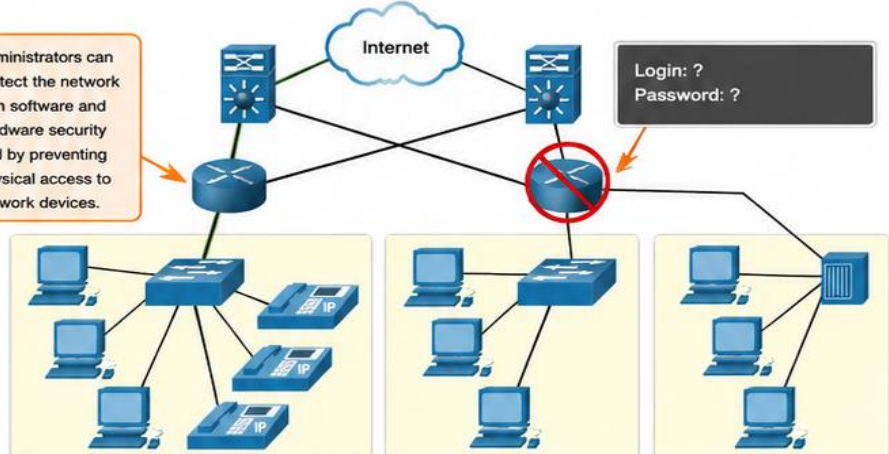
The network infrastructure, services, and the data contained on network-attached devices are crucial **personal and business assets**.

Network administrators must address two types of network security concerns:

- 1 Network infrastructure security
- 2 Information security

Securing the **network infrastructure** includes physically securing devices that provide network connectivity and preventing unauthorized access to the management software that resides on them, as shown in the figure.

Administrators can protect the network with software and hardware security and by preventing physical access to network devices.



Security measures protect the network from unauthorized access.

Network administrators must also protect the information contained within the packets being transmitted over the network, and the information stored on network-attached devices. In order to achieve the goals of network security, there are three primary requirements.



1 Confidentiality

Data confidentiality means that only the intended and authorized recipients can access and read data. It protects data from being seen by unauthorized users.



2 Integrity

Data integrity assures users that the information has not been altered in transmission, from origin to destination. It protects data from unauthorized changes.



3 Availability

Data availability assures users of timely and reliable access to data services for authorized users. It ensures that resources and information are available when needed.



In Short: Reliable networks are built strong, recover fast, grow easily, give better service, and stay secure!





NETWORK TRENDS (PART 1)



1.7.1 RECENT TRENDS



Networks are always changing. New technologies and end-user devices come to market, so businesses and consumers must continuously adjust to this ever-changing environment.



As a future IT professional (NetAcad student), you should know these important networking trends that affect organizations and consumers.

Key Networking Trends:

- ✓ Bring Your Own Device (BYOD)
- ✓ Online collaboration
- ✓ Video communications
- ✓ Cloud Computing



New devices, new ways of working,
new demands on networks.

1.7.2 BRING YOUR OWN DEVICE (BYOD)



Any device, for any content, in any manner – a major global trend.



It requires significant changes in how we use devices and safely connect them to networks. This is called Bring Your Own Device (BYOD).



BYOD gives users the **freedom** to use personal tools to access information and communicate across a business or campus network.



Because devices are cheaper and more advanced, employees and students may use **their own laptops, tablets, smartphones, notebooks, and e-readers** for personal use.



BYOD means **any device**, with any ownership, used anywhere.

Examples of BYOD Devices



Laptop



Tablet



Smartphone



Notebook



Smart Watch



E-reader

1.7.3 ONLINE COLLABORATION



People connect to the network not only to access data and applications, but also to **work together**.



Collaboration is defined as “the act of working with another or others on a joint project.”



Collaboration tools such as **Cisco WebEx** help employees, students, teachers, customers, and partners to instantly connect, interact, and achieve their goals together.

Benefits of Online Collaboration



Instantly
connect
with others



Work together
on projects
in real time



Share ideas
and solve
problems



Achieve
objectives
faster

1.7.4 VIDEO COMMUNICATIONS



Video is a critical part of networking for **communication, collaboration, and entertainment**.



Video calls can be made from anyone with an internet connection, **no matter where** they are located.



Video conferencing is a powerful tool for communicating **locally and globally**.



As organizations grow across different locations and cultures, video becomes a **critical requirement** for effective collaboration.



See. Hear. Collaborate. Anywhere.



NETWORK TRENDS (PART 2)

New technologies are shaping the way we live, work, and connect!



1.7.6 CLOUD COMPUTING



Cloud computing is one of the ways we access and store data.

It allows us to store personal files, even backup an entire drive on servers over the internet.



Applications like word processing and photo editing can be accessed using the cloud.



For businesses, cloud computing extends the capabilities of IT without heavy investment in:

- New infrastructure
- Training new personnel
- Licensing new software

Services are on-demand and delivered economically to any device, anywhere in the world without compromising security or function.



Cloud computing is possible because of **data centers**. Data centers are facilities used to house computer systems and components.

- A data center can be one room or a large building (one or more floors) or an entire warehouse-sized building.
- Very expensive to build and maintain.



For security, reliability, and fault tolerance, cloud providers store data in distributed data centers instead of in one place.

- Data is stored in multiple data centers at different locations.

FOUR PRIMARY TYPES OF CLOUDS

Cloud Type	Description
Public clouds	Cloud-based applications and services offered in a public cloud are made available to the general population. Services may be free or on a pay-per-use basis, such as online storage. The public cloud uses the internet to provide services.
Private clouds	Cloud-based applications and services offered in a private cloud are intended for a specific organization or entity (e.g., a government). It can be set up using the organization's private network (expensive to build and maintain) or managed by an outside organization with strict access security.
Hybrid clouds	A hybrid cloud is made up of two or more clouds (e.g., part private, part public), where each part remains a distinct object but are connected using a single architecture. Users can access various services based on their access rights.
Community clouds	A community cloud is for exclusive use by specific entities or organizations. Differences from public clouds are customized for the community's needs. Example: Healthcare organizations must follow rules (e.g., HIPAA) that require special authentication & confidentiality. It is used by organizations with similar needs and concerns.

1.7.9 VIDEO COMMUNICATIONS



Video is a critical part of networking for communication, collaboration, and entertainment.



Video calls can be made from anyone with an internet connection, **no matter where they are located**.



Video conferencing is a powerful tool for communicating **locally and globally**.



As organizations grow across different locations and cultures, video becomes a **critical requirement** for effective collaboration.



See. Hear. Collaborate. Anywhere.

1.7.7 BRING YOUR OWN DEVICE (BYOD)



The concept of any device, for any content, in any manner, is a major global trend.



It requires significant changes to how we use devices and safely connect them to networks. This is called Bring Your Own Device (BYOD).



BYOD gives users the **freedom** to use personal tools to access information and communicate across a business or campus network.



With cheaper and more advanced devices, employees and students may use their own **laptops, tablets, smartphones, notebooks, and e-readers** for personal use.



BYOD means any device, with any ownership, used anywhere.

Examples of BYOD Devices



1.7.8 ONLINE COLLABORATION



Individuals want to connect to the network not only for access to data applications, but also to collaborate with one another.



Collaboration is defined as "the act of working with another or others on a joint project."



Collaboration tools like **Cisco WebEx** give employees, students, teachers, customers, and partners a way to instantly connect, interact, and achieve their objectives.

Benefits of Online Collaboration



Instantly connect with others



Work together on projects in real time



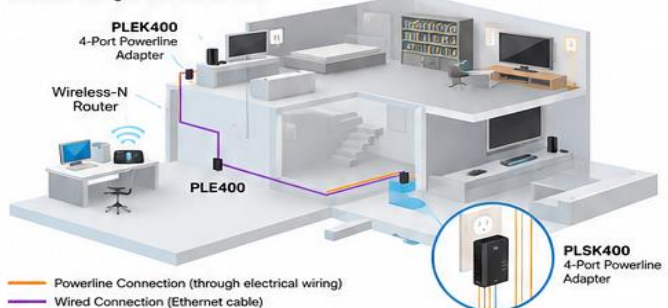
Share ideas and solve problems



Achieve objectives faster

1.7.10 POWERLINE NETWORKING

Powerline networking for home networks uses existing electrical wiring to connect devices.



Powerline Connection (through electrical wiring)
Wired Connection (Ethernet cable)



Using a standard powerline adapter, devices can connect to the LAN wherever there is an electrical outlet. No data cables need to be installed, and little to no additional electricity is used.



Powerline networking is very useful when wireless access points cannot reach all devices. It is not a substitute for dedicated cabling. However, it's an alternative when data cables or wireless communication are not possible or effective.

1.7.11 WIRELESS BROADBAND

In many areas where cable and DSL are not available, wireless may be used to connect to the internet.



Wireless Internet Service Provider (WISP)

- A WISP is an ISP that connects subscribers to a designated access point (hot spot) using wireless technologies similar to home WLANs.
- Common in rural areas where DSL or cable is not available.
- A separate transmission tower may be installed for the antenna (on water tower, radio tower, etc.).
- A small dish/antenna on the subscriber's roof (within range of the WISP transmitter).
- The access unit connects to the wired network inside the home.
- From the user's perspective, setup is similar to DSL/cable; the main difference is that connection from home to ISP is wireless.



Wireless Broadband Service

Provides high-speed wireless internet access to homes and businesses using technologies such as 4G/5G, satellite, or fixed wireless.



SECURITY THREATS & SOLUTIONS

Protecting Networks, Data, and Users



1.8.1 SECURITY THREATS

-  Network security is always a top priority because threat actors can try to access the personal information of thousands of customers.
-  Security is an integral part of computer networking, no matter if the network is in a home or a large company with thousands of users.
-  Security must consider the environment, available tools, and the quality of service (QoS) that users expect.
-  Securing a network uses protocols, technologies, devices, tools, and techniques to protect data and reduce threats. Threats can be external (from outside) or internal (from inside).
-  Many external threats today come from the internet. It is also important to consider internal threats.

COMMON EXTERNAL THREATS TO NETWORKS

-  **Viruses, worms, and Trojan horses** – These contain malicious software or code running on a user device.
-  **Spyware and adware** – These are types of software installed on a user's device that secretly collect information about the user.
-  **Zero-day attacks** – Also called zero-hour attacks, these occur on the first day that a vulnerability becomes known.
-  **Threat actor attacks** – A malicious person attacks user devices or network resources.
-  **Denial of service attacks** – These attacks slow down or crash applications and processes on a network device.
-  **Data interception and theft** – This attack captures private information from an organization's network.
-  **Identity theft** – This attack steals the login credentials of a user to access private data.

COMMON INTERNAL THREATS

-  Many studies show that most data breaches happen because of internal users. This can be due to:
 - Lost or stolen devices
 - Accidental misuse by employees
 - Malicious actions by employees

-  With BYOD and more connected devices, companies are more vulnerable. Security policies must address both external and internal threats.

1.8.2 SECURITY SOLUTIONS

-  No single solution can stop all threats. Security should be built in multiple layers, using more than one security solution. If one layer fails, other layers protect the network.

HOME OR SMALL OFFICE NETWORK SECURITY

Home network security is usually basic. It is applied at end devices, at the connection to the internet, and sometimes with ISP services.

Component	What It Does
 Antivirus and Antispyware	• Protect end devices (computers, phones, tablets, etc.) from getting infected by malicious software.
 Firewall Filtering	• Blocks unauthorized access in and out of the network. • Can be on the device (host-based firewall) or on the home router. • Prevents unwanted access from the outside internet.

CORPORATE NETWORK SECURITY

Corporate networks need stronger security because they handle more data and traffic. These components work together to monitor, control, and protect the network.

 Dedicated Firewall Systems	• Provide advanced firewall capabilities to filter large amounts of traffic with more granularity.
 Access Control Lists (ACL)	• Filter access and traffic forwarding based on IP addresses and applications.
 Intrusion Prevention Systems (IPS)	• Detect and stop fast-spreading threats, such as zero-day or zero-hour attacks.
 Virtual Private Networks (VPN)	• Provide secure access into an organization for remote workers.

★ KEY REMINDERS

- ✓ Security solutions must fit the environment, applications, and computing needs.
- ✓ Both home and business networks must protect data while still providing the quality of service (QoS) that users expect.
- ✓ Security solutions must adapt to new and changing network trends.
- ✓ Understanding switching and routing is the first step in studying network security threats and how to reduce them.

SUMMARY AT A GLANCE



SECURITY THREATS

Threats can come from outside the network (external threats) or inside the network (internal threats).

 Malware (viruses, worms, Trojan horses)	 Spyware & Adware	 Zero-day (Zero-hour) Attacks	 Threat Actor Attacks	 Denial of Service Attacks	 Data Interception & Theft	 Identity Theft
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INTERNAL THREATS

Caused by internal users due to lost/stolen devices, accidental misuse, or malicious employees.



SECURITY SOLUTIONS

Use multiple layers of security. If one layer fails, others protect the network.

 Home / Small Office	 Corporate Network
 Antivirus & Antispyware Protect devices from malicious software.	 Dedicated Firewall Systems – Advanced firewall to filter large traffic.
 Firewall Filtering Blocks unauthorized access in and out of the network.	 Access Control Lists (ACL) – Filter access and traffic based on IP & apps.
	 Intrusion Prevention Systems (IPS) – Detect and stop fast threats.
	 Virtual Private Networks (VPN) – Secure remote access for users.



Bottom Line: A secure network protects data, supports business, and builds trust.

